BDE & University/College Core Guide
NTU BSc Computer Science — Broadening & Deepening Electives, University Core & College Core (0–100)

NTU CS Curriculum Textbook Series
College of Computing and Data Science

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Chapter 1

What Is a Broadening & Deepening Elective?

> *"A computer scientist is someone who knows breadth — and chooses depth deliberately."* BDEs are NTU's formal mechanism for that choice: 20 AU of curated freedom outside your Programme Core and MPE.

From zero: no prior NTU jargon assumed. We define Academic Unit (AU), curriculum buckets, and every acronym (BDE, MPE, ICC, CSL, GER) from scratch. If you are in Y1 Week 1, start here.

Learning Outcomes

After this chapter you will be able to:

(L1) define AU, BDE, MPE, University Core, and College Core and state the 135 AU total;

(L2) explain *why* BDEs exist (breadth vs. depth) and where the 20 AU sits in the degree;

(L3) list the major BDE categories and give an example course for each;

(L4) distinguish GER-PE, GER-UE, and BDE counting rules at a high level;

(L5) locate the authoritative sources (Degree Audit, curriculum page, NTUMods) for your cohort.

1.1 Why BDEs Exist: Breadth, Depth, and Choice

Picture your degree as a fixed 135 AU "pie". About 60% is decided for you — University Core (17 AU) + College Core (16 AU) + Programme Core (47 AU) — and another 35 AU is semi-decided (MPE, from an SC3xxx/SC4xxx list). The remaining slice — about 15% — is *yours to design.* That slice is the Broadening & Deepening Electives (BDE) bucket: 20 AU (curriculum-page counting; the alternative Foundational-Core bucketing in the AY2024–25 PDF summary shows 21 AU for the same total — the 1 AU delta is a boundary shift between College Core and BDE, not extra workload; the 135 AU total is invariant).

"Broadening" means stepping outside Computing — business, psychology, language, art, sustainability. "Deepening" means going deeper within or adjacent to Computing but beyond the prescribed list — e.g., an extra CC000X-adjacent tech-and-society module or a Nanyang Business School analytics elective that complements your MPE. In practice most students blend both.

Definition 1.1 (Academic Unit (AU))**.** One AU = roughly 13 hours of total learning effort (lectures + tutorials + labs + self-study + assessment) over one 13-week semester. A 3 AU course ≈ 39 hours; a 4 AU course ≈ 52 hours. Your degree requires 135 AU for the single-degree BSc (Hons) in Computer Science (direct honours, 4 years). Variants — second major, double degree, University Scholars Programme — shift AU between buckets but keep the total fixed; confirm via Degree Audit.

Definition 1.2 (BDE vs. MPE — one sentence)**.** A *Major Prescribed Elective (MPE)* must be an SC3xxx or SC4xxx module from the CCDS MPE list (9 courses, ≥4 at 4000-level; FYP path includes SC4079). A *Broadening & Deepening Elective (BDE)* is any NTU-approved elective not already counted in University/College/Programme Core or MPE. The same course cannot count twice.

Remark 1.1 (Old terminology: GER-PE / GER-UE). Before AY2021, NTU used General Education Requirement (GER) labels: GER-Core, GER-PE (Prescribed Elective), GER-UE (Unrestricted Elective). UEs are essentially today's BDEs; PEs overlapped with today's MPE + some ICC. If a senior tells you "use your UEs for BDE", they mean the same bucket. This guide uses post-AY2021 language (BDE/MPE/ICC) throughout.

1.2 The 135 AU Map and Where BDEs Sit

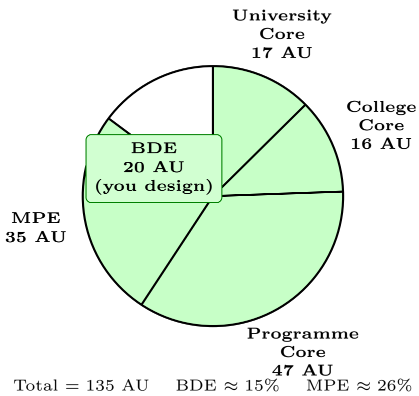


Figure 1.1: The 135 AU pie for single-degree CSC (curriculum-page bucketing). Two bucketings exist: curriculum page (ICC+CSL 17 + College 16 + Programme 47 + MPE 35 + BDE 20) and PDF Foundational-Core summary (Core 47 + MPE 35 + ICC 17 + F-Core 15 + BDE 21); both sum to 135. Degree Audit is authoritative for your cohort.

Table A.1 in the Appendix gives the same breakdown in tabular form with both bucketings side-by-side. Key intuition: BDEs are the *only* bucket where essentially any NTU course qualifies (subject to prerequisites and exclusion rules). They are also the bucket most often repurposed if you add a second major or minor — the second-major's required courses replace BDE AU one-for-one, shrinking free choice.

1.2.1 What Counts as a BDE?

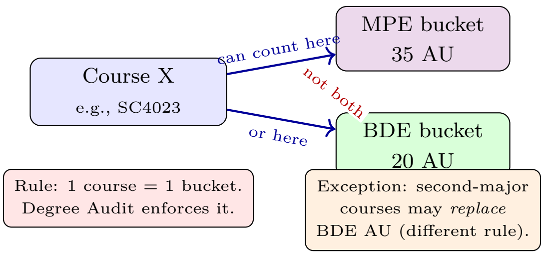
Formally, any NTU course that (i) carries AU, (ii) is not already satisfying University Core, College Core, Programme Core, or MPE for your degree, and (iii) is approved as a BDE in the NTU Course Catalogue is eligible. In practice:

- Language courses (LC5001 Chinese, LJ5001 Japanese, LK5001 Korean, LF5001 French, . . .) — 3–4 AU each.
- Business & management (AB1601 Organisational Behaviour, AB1202 Statistics & Analysis, AC1103 Accounting, . . .).
- Humanities, arts & social sciences (HP1000 Intro to Psychology, HP8001 Thinking & Reasoning, HAxxxx Art History, . . .).
- Science & sustainability outside core (CM8001 Energy, BS8003 Science of Food, . . .).
- Any SC-coded course *not* on the MPE list or not counted as Programme Core — rare, but possible (e.g., SC1006-level service modules for non-majors, if offered to CS).
- Entrepreneurship & innovation (ET9131 Entrepreneurship, . . .), physical education electives, and Nanyang Technopreneurship Centre modules.

What does *not* count: retaking a course you already passed (no new AU), a course whose content overlaps >50% with one you already took (exclusion list), or the same AU counted in two buckets (no double-counting — see §1.2.2).

1.2.2 Double-Counting: First Taste

The single most common BDE mistake is “double-counting”:



Chapter 2 catalogues the exact double-count traps CS students hit (e.g., listing an SC4xxx as both MPE and BDE).

1.3 BDE Categories as a Cloud

No category is compulsory. A “language + business + psychology” mix is as valid as “six entrepreneurship modules”. The only constraints are prerequisites (e.g., LJ5002 requires LJ5001), schedule clashes, and the double-count/exclusion rules above. Chapter 3 shows how to distribute these across four years.

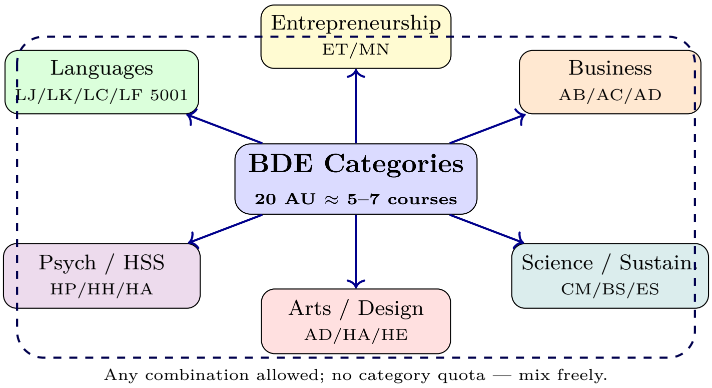


Figure 1.2: BDE category cloud for NTU CS. No category is compulsory and no quota per category exists; breadth is the point. Second-major students will see this cloud shrink as BDE AU is repurposed.

1.3.1 How to Verify for Your Cohort

NTU curricula drift. The figures in this guide reflect the AY2024–25 CCDS CSC study plan (135 AU invariant) and NTUMods data at time of writing. For *your* cohort:

1. **Degree Audit** (StudentLink → Academic → Degree Audit) — the single authoritative record. If this guide and Degree Audit disagree, Degree Audit wins.
2. **CCDS curriculum page** (ntu.edu.sg/computing → Discover CCDS → Curriculum) — bucket overview and study plan PDF.
3. **NTUMods / Course Catalogue** (ntu.edu.sg → Education → Courses) — per-course AU, prerequisites, exclusions, and whether a course is MPE-listed.

1.4 Chapter Notes

Curriculum buckets follow the CCDS BSc (Hons) in Computer Science programme structure (AY2024–25 study plan PDF and curriculum design page). AU definitions per NTU Academic Regulations. Historical GER terminology per NTU Curriculum Structure archives (pre-AY2021). Category examples drawn from NTU Course Catalogue listings at time of writing; offerings vary by semester.

1.5 Exercises

E1.1 **AU arithmetic.** You have completed 98 AU with 12 AU of BDE so far. How many BDE AU remain under the 20 AU bucket? If you add a second major requiring 28 AU that replaces BDE AU one-for-one, how many "free" BDE AU remain?

E1.2 **Double-count check.** You take SC4023 (AI for Software Engineering, 3 AU) listed as an MPE. Can you also count it toward BDE? What if SC4023 is *not* on the MPE list for your cohort but you still take it — does it count as BDE or neither?

E1.3 **Bucket invariant.** Explain why the two bucketings (curriculum page vs. PDF Foundational-Core summary) both sum to 135 AU despite showing BDE as 20 vs. 21.

E1.4 **Source triage.** A senior says "BDE is 21 AU now". Where do you verify, and which system is authoritative?

Chapter 2

Popular BDEs for CS Students

> *"Choose BDEs that compound — not just that are easy."* Breadth should make you more employable, more literate, and more interesting. Workload matters, but signal matters more.

How to read this chapter. We list the most-taken BDEs among CS cohorts, grouped by intent (career, language, GPA-management, curiosity), with workload notes and the double-count traps that catch students every semester.

Learning Outcomes

After this chapter you will be able to:

(L1) name 10+ popular BDEs for CS students and what each signals to employers/grad schools;

(L2) interpret a BDE workload table (contact hours, assessment mix, group-work load);

(L3) identify double-count and exclusion pitfalls (MPE vs. BDE, second-major overlap);

(L4) apply a 4-question filter to decide whether a candidate BDE is worth your AU.

2.1 Popular BDEs by Intent

Not every popular BDE is popular for the same reason. Group by *why* you are taking it:

Career-complement (Business / Finance / Analytics). CS + business literacy is the most common pairing. Recruiters for SWE, product, and quant-adjacent roles notice it.

- **AB1601 Organisational Behaviour & Design** (3 AU) — teams, motivation, org structure; heavy group project + presentation; moderate workload; no prerequisites.

- **AB1202 Statistics & Analysis** (3 AU) — overlaps conceptually with SC2000 but coded as Business; check exclusion with SC2000/MH2500 before enrolling.
- **AC1103 Accounting I / BF1101 Finance** — useful if targeting fintech; quantitative but memorisation-heavy on standards.
- **ET9131 Entrepreneurship I, BM2501 Business Analytics** — venture/innovation signal; project-heavy.

Language BDEs (global mobility). NTU's Centre for Modern Languages offers 3–4 AU sequences. CS students over-index here because language + technical skill is rare and durable.

- **LJ5001 Japanese Level 1** (4 AU) → LJ5002–5006 progression; **LK5001 Korean**, **LC5001 Chinese**, **LF5001 French**, **LH5001 Hindi** analogous.
- 4 AU for Level 1 (more contact hours), then 3 AU for higher levels. Entry by placement test if you have prior knowledge — do not enrol in Level 1 if you already speak the language; audit will flag it.

Human behaviour & society (HSS / Psychology). Design better products, manage teams, understand users.

- **HP1000 Introduction to Psychology** (3 AU) — perception, memory, social psych; exam + quizzes; no prerequisites; consistently high enrolment.
- **HP8001 Thinking & Reasoning / HP8101 Mind & Behaviour** (3 AU) — lighter, essay-based.
- **HH1001 What Is History? / HAxxxx Art History, HE9091 Principles of Economics** — HSS breadth; writing-intensive.

Science, sustainability & curiosity.

- **CM8001 Sustainable Energy, BS8003 Science of Cooking, ES9001 Sustainability** — popular as "lighter" BDEs; check workload — some have lab components.
- **GC0001 / EE8087 Cyber Ethics or Sustainability-adjacent ICC overflow** — if not already counting as ICC, may count as BDE; verify with Degree Audit (ICC vs. BDE boundary shifts by cohort).

Creative & wellness.

- Photography, film, music, creative writing (various HE/HA/AD codes), and Nanyang Technopreneurship Centre design modules. Workload is often portfolio/project-based — less exam stress, more time management.

Course (example)	AU	Contact	Exam	Group	Notes
HP1000 Intro to Psychology	3	3 h/wk	40–60%	Low	Quizzes + exam; no prereq.
AB1601 Org. Behaviour	3	3 h/wk	30–40%	High	Group project + presentation.
AB1202 Statistics & Analysis	3	3 h/wk	50–60%	Med	Check exclusion w. SC2000.
LJ5001 Japanese 1	4	4 h/wk	40%	Low	Daily vocab; placement test if prior.
LK5001 Korean 1	4	4 h/wk	40%	Low	Same pattern as LJ.
ET9131 Entrepreneurship	3	3 h/wk	20–30%	High	Pitch + business plan.
CM8001 Sustainable Energy	3	2–3 h/wk	50%	Low	Some semesters include lab.
HE9091 Economics	3	3 h/wk	50–60%	Low	Graph/mechanism heavy.
BM2501 Business Analytics	3	3 h/wk	40%	Med	Python/R overlap with CS — easy win.
HAxxx / ADxxx (arts/design)	3	2–3 h/wk	0–20%	Med	Portfolio-based; time-heavy.

Table 2.1: Illustrative workload snapshot for popular BDEs. Exact exam % and contact hours vary by semester/instructor — always check the course outline in NTU Course Catalogue before bidding. "Group" = group-project load.

2.2 Workload & Assessment Table

Remark 2.1 (Reading the table). A "Low group, 50% exam" BDE (HP1000, HE9091) suits a semester already heavy on group-heavy core courses (SC2006, SC3010). A "High group" BDE (AB1601, ET9131) is better in a semester with exam-heavy cores. Balance your semester, not just the single course.

2.3 The Double-Count & Exclusion Flowchart

The same course cannot satisfy two buckets. Use this decision tree *before* you bid:

Three concrete traps.

1. **SC-coded MPE trap.** You take SC4023 (MPE-listed) thinking "one more SC4xxx as BDE". It counts as MPE only. If your MPE bucket is already full (9 courses + FYP), the extra SC4xxx may still count as BDE — but only if you *declare* it unused for MPE and your Degree Audit agrees. Do not assume; declare explicitly.
2. **AB1202 ↔ SC2000 exclusion.** Some Business statistics courses exclude SC2000/MH2500. You passed SC2000 in Y1. AB1202 may be blocked. Check the "Mutually exclusive" line in the catalogue, not just prerequisites.
3. **Language placement.** You speak conversational Korean and enrol in LK5001 (Beginner). The Centre

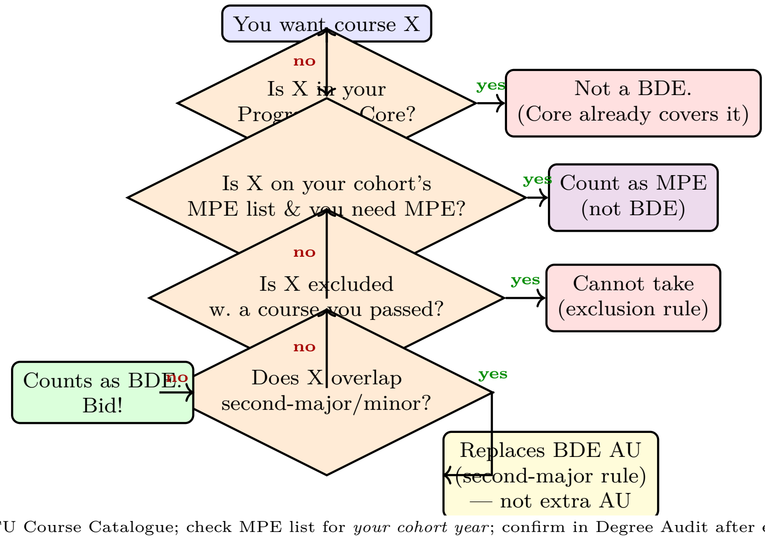
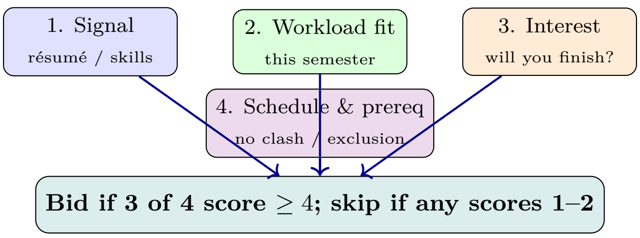


Figure 2.1: Should this course count as BDE? Follow the tree. When in doubt, email your academic advisor *before* bidding — STARS swaps cost time and may lose your slot.

for Modern Languages may require a placement test and reassign you to LK5003. Taking LK5001 anyway risks AU not counting.

2.4 A 4-Question Filter for Any Candidate BDE

Before you bid, score the candidate 1–5 on each:



A BDE that scores 5 on interest but 1 on workload fit for *this* semester should wait — the catalogue repeats most BDEs every semester or year.

2.5 Chapter Notes

Course examples and workload patterns synthesised from NTU Course Catalogue entries, NTUMods, and CCDS student forum reports at time of writing; offerings, AUs, and assessment mixes vary by semester and instructor. Always verify prerequisites, exclusions, and MPE-list status for your cohort year before bidding.

2.6 Exercises

E2.1 **Workload balancing.** You are taking SC2006 (group-heavy) + SC2005 (exam-heavy) + SC2203 (proof-heavy) in one semester and want one BDE. Which BDE from Table 2.1 balances best, and why?

E2.2 **Exclusion check.** You passed SC2000. A friend says AB1202 is a free 3 AU BDE. What do you check before bidding, and where?

E2.3 **MPE vs. BDE declaration.** You have 8 MPE courses done and need one more MPE at 4000-level. You also want SC4023 as a BDE. Can SC4023 satisfy both? What must you do in Degree Audit if your MPE bucket is already full?

E2.4 **Language placement.** You grew up speaking Japanese at home. Should you enrol in LJ5001? What is the correct first step?

Chapter 3

Planning Your BDEs Across Four Years

"A degree is a schedule, not just a checklist." Where you place BDEs matters as much as which BDEs you pick. A good plan keeps semesters balanced, reserves AU for late surprises, and never forces you into a 24 AU overload to graduate.

From zero: we define Degree Audit, STARS, overload, and S/U before using them. No prior NTU registration experience assumed.

Learning Outcomes

After this chapter you will be able to:

(L1) interpret a four-year study plan and place BDEs sensibly across semesters;

(L2) read Degree Audit correctly (Satisfied / In-Progress / Outstanding);

(L3) state overload rules, normal semester load, and when overloading is (not) advisable;

(L4) plan around internship (SC3920) and FYP/MPE-heavy final year;

(L5) handle STARS bidding, waitlists, and late BDE swaps without panic.

3.1 A Sample Four-Year BDE Distribution

The programme core is front-loaded: Y1–Y2 carry 15–18 AU of compulsory SC/MH modules per semester. Y3 S2 is dominated by the 10 AU Professional Internship (SC3920). Y4 is MPE + FYP (or 3 extra MPE). BDEs fill the gaps.

Remark 3.1 (Why not front-load all BDEs?). Three reasons: (i) prerequisites — some desirable BDEs (e.g., BM2501 Business Analytics) expect Y2 standing; (ii) timetable — Y1 cores have fixed lecture slots that clash with popular BDEs; (iii) GPA smoothing — spreading BDEs lets you balance exam-heavy and project-heavy semesters (see Table 2.1 in Ch. 2).

Year / Sem	Core load (approx.)	BDE slot	BDE AU
Y1 S1	15–16 AU (SC1003/04/05, MH1812, ICC)	1 BDE if timetable allows (e.g., HP1000)	0–3
Y1 S2	16–18 AU (SC1006/07/08, SC2000/02)	0–1 BDE (light if core is heavy)	0–3
Y2 S1	15 AU (SC2001/05/06, SC2203/08)	1 BDE (business or lang. 1)	3
Y2 S2	12–15 AU (SC2207 + MPE start)	1 BDE (lang. 2 or HSS)	3
Y3 S1	14–16 AU (SC3099 4 AU + MPEs)	1 BDE (entrepreneurship / arts)	3
Y3 S2	**10 AU internship** + 1 ICC/MPE	0 BDE (internship is full-time)	0
Y4 S1	MPE/FYP 8–12 AU	1–2 BDEs (clear remaining)	3–6
Y4 S2	MPE/FYP 8–12 AU	1 BDE (final buffer)	3
		Total BDE	**18–21** → **20** target

Table 3.1: Sample BDE distribution (single-degree, no second major, FYP path). Adjust per your actual core load and internship timing. The 20 AU target is curriculum-page counting; PDF F-Core bucketing shows 21 AU for the same courses.

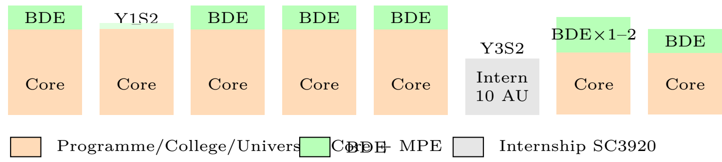
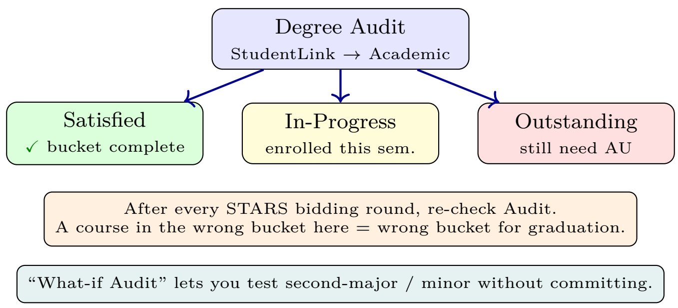


Figure 3.1: Schematic BDE placement across eight semesters (Y3 S2 internship has no BDE). Heights are schematic; actual AU per semester is in Table 3.1. Leave at least one BDE buffer for Y4 in case of timetable clashes or failed bids.

Buffer rule. Plan to finish BDEs by Y4 S1, leaving Y4 S2 as buffer. Students who postpone 6+ AU of BDE to the very last semester risk a timetable clash forcing an overload or delayed graduation.

3.2 Degree Audit: The Single Source of Truth

Definition 3.1 (Degree Audit)**.** Degree Audit is the official, personalised checklist in StudentLink (Academic → Degree Audit) that maps every course you have taken, are taking, and plan to take into the correct AU bucket (University Core, College Core, Programme Core, MPE, BDE). It shows Satisfied (green), In-Progress (yellow), and Outstanding (red) for each requirement. If Degree Audit says a requirement is unsatisfied, it *is* unsatisfied — regardless of what any PDF or senior says.



How to use it for BDEs.

1. After STARS enrolment each semester, open Degree Audit and confirm the new course landed in the BDE bucket (not "Unallocated" or MPE).
2. If a BDE appears under "Unallocated" (e.g., an SC-coded extra that the system thinks is MPE), email your programme coordinator to reallocate — do not leave it until Y4.
3. Use "What-if Audit" to simulate adding a second major or minor and see how many BDE AU it consumes.

3.3 Semester Load, Overload, and S/U

Definition 3.2 (Semester load & overload)**.** NTU defines a *normal load* of roughly 16–20 AU per semester (5–6 courses). The *maximum load without special approval* is typically 20–21 AU; beyond that is *overload* and requires academic approval (good GPA, advisor sign-off). The absolute cap per semester is usually 24 AU. Internship semester (SC3920, 10 AU) is an exception — you take fewer or no additional courses alongside it.

Definition 3.3 (S/U option)**.** Satisfactory/Unsatisfactory (S/U) lets you take a course without it affecting GPA (S = pass, U = fail, neither enters GPA). At NTU, S/U eligibility is restricted — typically only BDEs and some ICC courses qualify, and only up to a fixed AU cap (check Academic Regulations for your cohort). Programme Core and MPE generally *cannot* be S/U'd.

Rule of thumb	Why
Keep 17–20 AU/semester	Sustainable; leaves time for CCAs, projects, interviews.
Avoid 22+ AU unless GPA ≥ 4.0	Overload approval is not guaranteed; burnout risk is real.
Do not S/U a BDE you need for signal	Employers/grad schools see S, not the grade; S hides a good grade too.
Reserve 3–6 AU BDE for Y4	Buffer for timetable clashes or a failed/dropped BDE.
Check STARS vacancy before planning	Popular BDEs (HP1000, AB1601) fill in minutes; have backups.

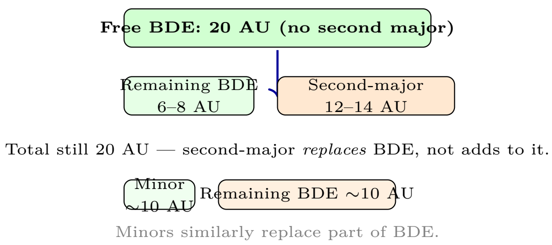
Table 3.2: Overload and planning heuristics for CS students.

STARS bidding tips.

- STARS (Student Automated Registration System) allocates courses by vacancy and priority (programme cores first, then MPE, then BDE). BDEs are lowest priority — vacancies are tightest.
- Have 2–3 backup BDEs per slot. If your first choice is full, bid the backup immediately; do not wait for add/drop hoping a slot opens.
- Add/drop week (first week of semester) is your last chance to swap without penalty — attend both the enrolled and desired BDE's first class if you are waitlisted.

3.4 Second Major, Minor, and BDE Trade-offs

A second major (e.g., Business, Mathematics) or minor consumes AU that would otherwise be BDE. The "What-if Audit" shows the exact trade:



If you declare a second major, your "free BDE" shrinks — but the second-major courses themselves broaden you, so the spirit of BDE is preserved. Decide by Y2 S1; late switches waste AU.

3.5 Chapter Notes

Load and overload rules per NTU Academic Regulations and CCDS programme structure (AY2024–25). STARS procedures per NTU StudentLink documentation. S/U rules vary by cohort and programme — confirm in Degree Audit and Academic Regulations.

3.6 Exercises

E3.1 **Semester balance.** You must take SC2006 (group-heavy, 20 h/wk with project) and SC2005 (exam-heavy) next semester. You also need 3 AU of BDE. Which BDE workload profile (exam-heavy vs. group-heavy) balances better, and why?

E3.2 **Audit triage.** After STARS, Degree Audit shows your new BDE under "Unallocated" instead of BDE. What happened, and what do you do?

E3.3 **Buffer planning.** A friend plans to finish all 20 AU of BDE by Y3 S1 and leave Y4 with only MPE/FYP. Is this risky or safe? Give two failure modes if it goes wrong.

E3.4 **Second-major math.** A second major requires 30 AU. How many "free" BDE AU remain under the 20 AU bucket if you declare it? What Degree Audit feature lets you test this before committing?

Chapter 4

College Core: The Professional Spine

"College Core is the bridge from classroom to workplace." These are not BDEs — they are compulsory professional modules owned by CCDS. Confusing College Core with BDE is a common audit error; this chapter makes the distinction crisp.

From zero: College Core defined. College Core = the *common professional spine* for every CCDS BSc student (CS, AI, Data Science). It is *not* optional, *not* BDE, and *not* MPE. We cover each component—purpose, AU, timing, and assessment—then contrast SC3099 vs. SC4079.

Learning Outcomes

After this chapter you will be able to:

(L1) list the College Core components, AU, and typical semester for each;

(L2) explain the purpose of SC1003, HW0288, SC3920, and the SC3099/SC4079 projects;

(L3) distinguish SC3099 Capstone (Programme Core, Y3 S1) from SC4079 FYP (MPE, Y4);

(L4) state prerequisites, grading, and workload expectations for each College Core course;

(L5) avoid the common mistake of counting a College Core course as BDE.

4.1 What Is College Core?

Definition 4.1 (College Core (Professional Series))**.** College Core is a CCDS-level requirement of 16 AU (curriculum-page bucketing; the alternative PDF Foundational-Core summary shows 15 AU for the same courses — the 1 AU delta is a boundary shift with BDE, total 135 AU invariant). Every CCDS BSc student — CS, AI, Data Science — takes the same College Core spine. In the curriculum-page bucketing it comprises:

Course	AU	When	Purpose
SC1003 Intro to Computational Thinking & Programming	3	Y1 S1	Foundational programming & problem solving
HW0288 Engineering Communication	2	Y3	Professional writing, presentation, teamwork
SC3920 Professional Internship	10	Y3 S2	Full-time industry attachment (graded)

Plus 1 AU institutional rounding / bridging that appears as BDE in the PDF bucketing.

SC3099 (Capstone, 4 AU, Y3 S1) and SC4079 (FYP, 8 AU, Y4) are *narratively* part of the professional spine but bucketed differently in the official summaries: SC3099 is Programme Core (Core 47 AU in the PDF) and SC4079 is MPE (see below). Table A.1 in the Appendix shows both bucketings side-by-side.

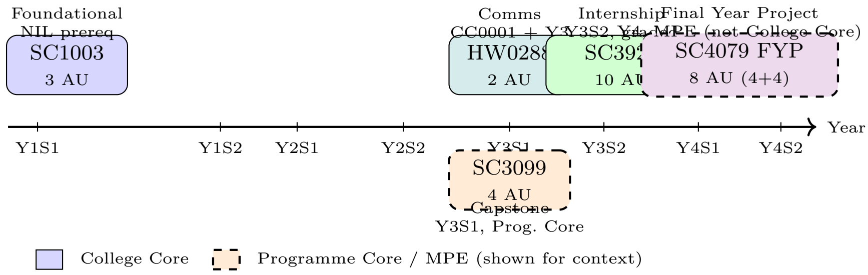


Figure 4.1: College Core timeline (solid) with SC3099/SC4079 shown dashed for context. Y3 S2 internship dominates that semester — no BDE alongside it.

Remark 4.1 (Bucketing confusion). If you compare the curriculum page (College Core 16 AU) with the PDF summary (Foundational Core 15 AU + Core 47 AU that includes SC3099), the 1 AU difference is the same BDE-boundary shift seen in Ch. 1. Degree Audit reconciles both views; the course list is identical.

4.2 The Three College Core Courses in Detail

4.2.1 SC1003 Introduction to Computational Thinking & Programming (3 AU)

When & prerequisites. Y1 S1, no prerequisites (NIL). Co-requisite for SC1004 (Linear Algebra for Computing).

What it is. Your first programming course: problem decomposition, algorithmic thinking, Python (or cohort language), basic data structures, testing, and debugging. It also co-introduces the mathematical maturity (sets, logic) that MH1812 and SC2001 will formalise.

Workload. 3 h lecture + 1 h tutorial + 2 h lab per week is typical; continuous assessment (labs, quizzes, mini-project) plus a final exam. Pass SC1003 early — it unlocks SC1007, SC1008, SC2002, and indirectly most of Y2.

Why it is College Core, not BDE. It is foundational for every CCDS programme; without it, downstream cores have no programming base. Counting it as BDE would undercount your core AU.

4.2.2 HW0288 Engineering Communication (2 AU)

When & prerequisites. Y3, prerequisites CC0001 (Inquiry & Communication) *and* Year 3 standing. Offered by the School of Humanities (HSS) for CCDS.

What it is. Professional communication: technical writing (reports, proposals), oral presentation, visual communication, and teamwork dynamics. Assessment is heavily continuous — presentations, group report, peer review — with little or no final exam.

Why it matters for CS. Code is read more than written; so are design docs, incident reports, and product specs. HW0288 is the only compulsory course that grades your ability to explain technical work to mixed audiences. Recruiters who ask for "strong communication skills" are asking for what this course practises.

Not a BDE. HW0288 is compulsory College Core; you cannot S/U it as a BDE or replace it with another HSS course.

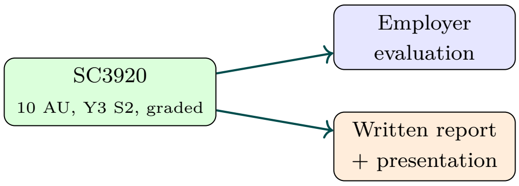
4.2.3 SC3920 Professional Internship (10 AU, Graded)

When & prerequisites. Y3 S2, Year 3 standing. Full-time, typically 20–24 weeks.

What it is. A graded, credit-bearing attachment in industry (or approved research/overseas internship). You are assessed on work performance (employer evaluation), a written report, and often a presentation. The grade counts toward GPA and honours classification.

Key rules.

- **Graded, not Pass/Fail.** Unlike some universities' internships, SC3920 carries a letter grade that affects GPA. Treat it as a 10 AU course, not a "break".
- **Full-time commitment.** Do not plan BDEs or heavy MPEs alongside SC3920. The official guidance is to take at most one additional small course (often an ICC or online module) if your employer schedule permits; many students take none.
- **Timing alternatives.** A small number of students intern in Y3 S1 or Y4 by arrangement; this shifts SC3099/MPE timing. Confirm with CCDS Undergraduate Office before moving it.
- **Overseas / research internships** are possible but require prior approval and may have different deliverables.



Both components graded; GPA-relevant.

4.3 SC3099 vs. SC4079: Two Different Projects

This is the most confused pair in the programme. They are different courses, different years, different buckets, and different honours implications.

	SC3099 Capstone Project	**SC4079 Final Year Project (FYP)**
AU	4 AU	8 AU (4 + 4 across Y4 S1 & S2)
Bucket	Programme Core (Core 47)	MPE (counts toward 35 AU MPE)
When	Y3 S1, Year 3 standing	Y4 S1–S2, Year 4 standing
Team	Team-based (3–5 students)	Individual (supervised by faculty)
Deliverable	Integrated system / product + report + demo	Research or engineering thesis + defence
Grading	Team + individual components	Individual; supervisor + examiner
Honours	Required for all	Required for Highest Distinction; alternative is 3 extra MPE (see below)

Table 4.1: SC3099 vs. SC4079 at a glance. Do not confuse them; Degree Audit lists them separately.

Remark 4.2 (FYP-or-3-MPE rule (PDF Notes 2–3))*.* To satisfy the 35 AU MPE requirement you have two paths:

- **FYP path (standard):** 27 AU of MPE courses (9 courses × 3 AU, with ≥4 at 4000-level) + SC4079 (8 AU) = 35 AU MPE. Required for Highest Distinction eligibility.
- **Non-FYP path:** 27 AU of MPE courses + 3 *additional* MPE courses (9 AU) = 36 AU MPE. Total degree becomes 136 AU unless BDE is adjusted per Degree Audit; confirm with your advisor. Highest Distinction is not attainable on this path.

Either way, SC3099 is *always* required as Programme Core — it is not interchangeable with SC4079.

4.4 Chapter Notes

Course details per CCDS BSc (Hons) in Computer Science study plan (AY2024–25), NTU Course Catalogue entries for SC1003/HW0288/SC3920/SC3099/SC4079, and PDF Notes 1–3 on MPE/FYP. Assessment mixes

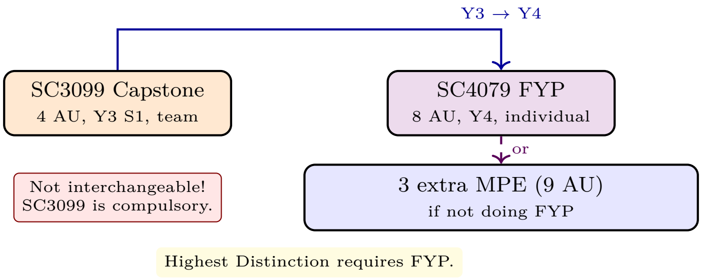


Figure 4.2: SC3099 and SC4079 are sequential, not alternative. The *alternative* is FYP vs. 3 extra MPE for the MPE bucket.

vary by semester; confirm in the course outline. Internship duration and grading per CCDS Undergraduate Office guidelines at time of writing.

4.5 Exercises

E4.1 **Bucket check.** A student lists SC1003 as a BDE in their plan. Why is this wrong, and which bucket does it belong to?

E4.2 **Capstone vs. FYP.** State two differences between SC3099 and SC4079 (AU, bucket, timing, or team structure).

E4.3 **Internship planning.** Can you take a 3 AU BDE alongside SC3920 in Y3 S2? What is the recommended load that semester?

E4.4 **Honours rule.** You want to graduate with Highest Distinction but prefer not to do an FYP. Is this possible? What does the programme require?

Chapter 5

University Core: ICC & Career Skills

"NTU wants every graduate — engineer or historian — to be digitally literate, ethically grounded, and career-ready." University Core is that common contract. It is not BDE, not College Core, and not optional.

From zero: University Core defined. University Core = the *NTU-wide* compulsory spine for every undergraduate (all colleges, all programmes). It comprises the Interdisciplinary Collaborative Core (ICC, CC0001–CC0007) plus Career & Innovative Enterprise for the Future World (ML0004, CSL). Total: 17 AU. We cover each course's purpose, AU, and how ICC differs from BDE/College Core.

Learning Outcomes

After this chapter you will be able to:

(L1) list every University Core component (CC0001–CC0007 + ML0004), AU, and typical year;

(L2) explain the ICC's five learning pillars and why CS students take ICC alongside computing cores;

(L3) distinguish University Core from College Core and BDE (no double-counting);

(L4) describe ML0004 (CSL) and its career-preparation purpose;

(L5) locate ICC/CSL in your timetable and Degree Audit.

5.1 What Is University Core?

Definition 5.1 (University Core (ICC + CSL) — 17 AU)**.** University Core is NTU-wide: every undergraduate, from CCDS to HSS to Business, takes the same spine (with minor college-specific variants in delivery). It has two parts:

- **Interdisciplinary Collaborative Core (ICC)** — CC0001, CC0002, CC0003, CC0005, CC0006, CC0007 (15 AU combined).

- **Career & Innovative Enterprise for the Future World (CSL)** — ML0004 (2 AU).

Total: 17 AU. University Core is *not* BDE and *not* College Core; a course counted as ICC cannot also count as BDE. Table A.1 in the Appendix shows the full 135 AU map.

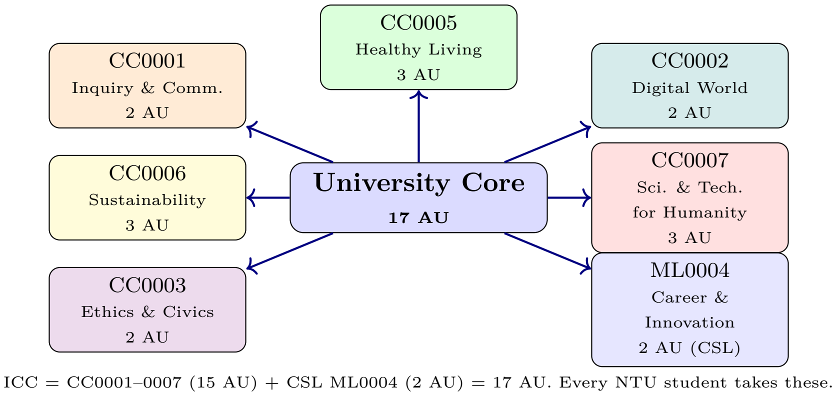


Figure 5.1: University Core flower: every petal is compulsory. No ICC course can be double-counted as BDE.

5.2 The ICC Courses in Detail

5.2.1 CC0001 Inquiry & Communication in an Interdisciplinary World (2 AU)

Purpose. Academic inquiry, critical reading, argument construction, and written/oral communication across disciplines. You learn to ask good questions, evaluate evidence, and present an argument — not just to write essays.

When. Y1 S1 for most CCDS cohorts. Prerequisite for HW0288 (College Core) and for CC0003 in some pathways.

Why CS students need it. Design docs, RFCs, incident reports, and paper reviews all require inquiry + communication. CC0001 is the foundation; HW0288 later specialises it for engineering contexts.

5.2.2 CC0002 Navigating the Digital World (2 AU)

Purpose. Digital literacy for every NTU graduate: data, AI, cybersecurity, and digital ethics at a conceptual level — not coding, but understanding the digital society you will build systems for.

For CS students. Much of CC0002 will feel familiar (you go deeper in SC1005/SC2006/SC3010). Treat it as the "explain it to a non-CS friend" version — valuable precisely because you will need to explain CS to non-CS stakeholders.

5.2.3 CC0003 Ethics & Civics in a Multicultural World (2 AU)

Purpose. Ethical reasoning, civic responsibility, and multicultural competence in Singapore and globally. Case studies on technology ethics, inequality, and governance.

For CS students. Directly relevant to AI ethics, algorithmic bias, data privacy (PDPA/GDPR), and responsible deployment — themes that recur in SC3000, SC3010, and your future professional practice.

5.2.4 CC0005 Healthy Living & Wellbeing (3 AU)

Purpose. Physical health, mental wellbeing, and resilience. Includes pathways HW0001/HW0002 for students who need bridging.

Format. Often blended — lectures + activity-based components (fitness, mindfulness, health literacy). Assessment is typically quizzes + reflection + participation rather than a heavy exam.

5.2.5 CC0006 Sustainability: Society, Economy & Environment (3 AU)

Purpose. Climate, sustainability, circular economy, and the UN Sustainable Development Goals. Interdisciplinary by design — you collaborate with students from other colleges.

For CS students. Sustainable computing (energy-efficient systems, green AI, e-waste) connects directly. Several MPE and BDE sustainability modules (ES9001, CM8001) deepen this — but CC0006 itself is University Core, not BDE.

5.2.6 CC0007 Science & Technology for Humanity (3 AU)

Purpose. How science and technology shape humanity — history of technology, innovation, science communication, and societal impact.

For CS students. The "why should society trust your system?" course. Useful framing for FYP/Capstone impact sections and for communicating technical work to non-technical audiences.

5.3 ML0004 Career & Innovative Enterprise for the Future World (2 AU, CSL)

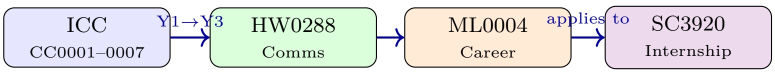
Definition 5.2 (CSL: Career Skills & Language? No — Career & Innovative Enterprise)**.** ML0004 is the *Career & Innovative Enterprise for the Future World* module (2 AU), classified as Career Skills & Leadership (CSL) in some summaries. It is distinct from the Centre for Modern Languages (also "language" in other contexts)

— same acronym space, different meaning. In University Core counting, ML0004 is the CSL component that brings University Core from 15 AU (ICC alone) to 17 AU.

Purpose. Career readiness: self-awareness, industry exploration, résumé and interview skills, networking, innovation and enterprise mindset. Delivered as workshops, industry talks, and reflective assignments rather than lectures + exam.

When. Typically Y2–Y3; some cohorts take it in Y1 S2. Check your study plan — timing shifts by cohort.

Why it matters. It is the only compulsory course explicitly about *getting and succeeding in* the internship (SC3920) and first job. Skipping its résumé/interview preparation before Y3 S2 internship applications is a common regret.



University Core (ICC + CSL) → College Core (Comms + Internship) — a deliberate pipeline.

Do ML0004 *before* internship applications (Y3 S1 latest).

Figure 5.2: From University Core to workplace: ICC builds breadth, HW0288 hones engineering communication, ML0004 prepares the résumé, SC3920 tests it in industry.

5.4 University Core vs. College Core vs. BDE

Bucket	AU	Owner	Can you choose?
University Core (ICC+CSL)	17	NTU (all colleges)	No — every course fixed
College Core	16	CCDS	No — SC1003/HW0288/SC3920 fixed
Programme Core	47	CCDS-CS	No — SC1004–SC3099 fixed
MPE	35	CCDS-CS	Semi — choose 9 from SC3xxx/4xxx list
BDE	20	You	Yes — any approved NTU course

AU per curriculum-page bucketing; PDF F-Core bucketing shows College 15 + BDE 21 for same courses. Total 135 invariant; Degree Audit authoritative.

Remark 5.1 (No double-counting across cores). A course counted as University Core (e.g., CC0002) cannot also satisfy College Core or BDE, even if its content overlaps with a CS-adjacent BDE. Degree Audit enforces one-bucket-per-course. If you take an extra CC-coded course beyond the required ICC, the *extra* one may count as BDE — but the required ICC ones never do.

5.5 Chapter Notes

University Core structure per NTU Interdisciplinary Collaborative Core (ICC) programme pages and NTU Academic Regulations (AY2024–25). Course purposes summarised from NTU Course Catalogue entries for CC0001–CC0007 and ML0004 at time of writing; titles and AUs confirmed against CCDS BSc (Hons) in Computer Science study plan PDF. Offerings and delivery mode vary by semester and cohort; Degree Audit is authoritative.

5.6 Exercises

E5.1 **AU check.** List the AU for each University Core course and verify the total is 17. Which component is CSL, and how many AU does it carry?

E5.2 **Bucket confusion.** A friend says "I will count CC0002 as a BDE because it is about digital literacy and I am a CS student". Why is this incorrect?

E5.3 **Career pipeline.** Arrange CC0001, HW0288, ML0004, and SC3920 in the order you should take them, and explain the dependency.

E5.4 **Degree Audit.** After Y1, Degree Audit shows CC0005 as "Outstanding" though you passed a sustainability BDE (ES9001). Why does ES9001 not satisfy CC0006, and what must you do?

Appendix A

Appendix: AU Summary Tables

This appendix gives the complete AU accounting for the single-degree BSc (Hons) in Computer Science (AY2024–25) in both official bucketings. The course list is identical; only the bucket labels differ by 1 AU at the College Core / BDE boundary. Degree Audit is authoritative for your cohort.

A.1 Full 135 AU Map — Both Bucketings

Component	AU		Notes
	Curriculum page	PDF F-Core summary	
University Core (ICC + CSL)	17	17	CC0001 (2) + CC0002 (2) + CC0003 (2) + CC0005 (3) + CC0006 (3) + CC0007 (3) + ML0004 (2)
College Core (Professional Series)	16	15 (F-Core)	SC1003 (3) + HW0288 (2) + SC3920 (10) + 1 AU rounding/bridging
Programme Core	47	47 (Core)	SC1004 (4), SC1005 (3), MH1812 (3), SC1006 (3), SC1007 (3), SC1008 (3), SC2000 (3), SC2002 (3), SC2001 (3), SC2005 (3), SC2203 (3), SC2006 (3), SC2008 (3), SC2207 (3), SC3099 (4)
Major Prescribed Electives (MPE)	35	35	FYP path: 9 courses $\times$ 3 + SC4079 (8) = 35; non-FYP: 12 $\times$ 3 = 36 (see text)
Broadening & Deepening Electives (BDE)	20	21	Any approved NTU courses not counted elsewhere
Total	**135**	**135**	Invariant; bucket labels shift by 1 AU at College/BDE boundary

Table A.1: 135 AU accounting: curriculum-page vs. PDF Foundational-Core bucketing. Both sum to 135; the 1 AU delta is a boundary shift, not extra workload.

Course	AU	Title / purpose
CC0001	2	Inquiry & Communication in an Interdisciplinary World
CC0002	2	Navigating the Digital World
CC0003	2	Ethics & Civics in a Multicultural World
CC0005	3	Healthy Living & Wellbeing
CC0006	3	Sustainability: Society, Economy & Environment
CC0007	3	Science & Technology for Humanity
ML0004	2	Career & Innovative Enterprise for the Future World (CSL)
Total	17	

Course	AU	When	Notes
SC1003	3	Y1 S1	Intro to Computational Thinking & Programming (NIL prereq)
HW0288	2	Y3	Engineering Communication (prereq CC0001 + Y3 standing)
SC3920	10	Y3 S2	Professional Internship (graded, Year 3 standing)

Plus 1 AU bridging/rounding that appears under BDE in PDF summary.

A.2 University Core Detail (17 AU)

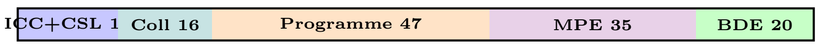
A.3 College Core Detail (16 AU curriculum page / 15 AU PDF F-Core)

A.4 Programme Core Detail (47 AU)

A.5 MPE Detail (35 AU FYP path / 36 AU non-FYP)

Note: SC4079 is itself an MPE (not College Core). The FYP-or-3-MPE choice is governed by PDF Notes 2–3: Highest Distinction requires FYP. Confirm via Degree Audit; the non-FYP total of 136 AU may be adjusted via BDE/overload per audit.

A.6 BDE Categories (20 AU / 21 AU)



135 AU total (curriculum-page bucketing). PDF F-Core bucketing: College 15 + BDE 21, same total.

Course	AU	Semester	Focus
SC1004	4	Y1 S1	Linear Algebra for Computing
SC1005	3	Y1 S1	Digital Logic
MH1812	3	Y1 S1	Discrete Mathematics (cohort-dependent; SC1123/24 alt.)
SC1006	3	Y1 S2	Computer Organisation & Architecture
SC1007	3	Y1 S2	Data Structures & Algorithms
SC1008	3	Y1 S2	C/C++ Programming
SC2000	3	Y1 S2	Probability & Statistics for Computing
SC2002	3	Y1 S2	Object-Oriented Design & Programming
SC2001	3	Y2 S1	Algorithm Design & Analysis
SC2005	3	Y2 S1	Operating Systems
SC2203	3	Y2 S1	Automata, Computability & Complexity
SC2006	3	Y2 S1	Software Engineering
SC2008	3	Y2 S1	Computer Network
SC2207	3	Y2 S2	Introduction to Databases
SC3099	4	Y3 S1	Capstone Project (team, Programme Core)
Total	47		

Verification Checklist for Your Cohort

1. Open Degree Audit (StudentLink → Academic → Degree Audit) and confirm each bucket shows Satisfied/In-Progress for your actual courses.
2. For any BDE, verify in NTU Course Catalogue: AU, prerequisites, "Mutually exclusive" line, and MPE-list status for your cohort year.
3. If Degree Audit and this guide disagree on AU bucketing, Degree Audit wins. If this guide and a senior disagree, Degree Audit wins.

Path	AU	Composition
FYP path (standard)	35	9 MPE courses × 3 AU (any SC3xxx/4xxx, ≥4 at 4000-level) + SC4079 FYP (8 AU: 4+4 over Y4)
Non-FYP path	36	12 MPE courses × 3 AU (any SC3xxx/4xxx, ≥4 at 4000-level); no FYP; Highest Distinction not eligible

Category	Example codes	Notes
Languages	LJ5001 (Japanese), LK5001 (Korean), LC5001 (Chinese), LF5001 (French)	4 AU Level 1, 3 AU higher; placement test if prior knowledge
Business & management	AB1601, AB1202, AC1103, BF1101, BM2501	Check exclusions (e.g., AB1202 vs. SC2000)
HSS / Psychology	HP1000, HP8001, HH1001, HE9091	No prerequisites for most; writing- or exam-heavy
Science / sustainability	CM8001, BS8003, ES9001	Some have labs; not double-countable with ICC
Entrepreneurship	ET9131, ET9132, MN3001	Project/pitch-heavy
Arts / design	HAxxxx, ADxxxx, HE-coded creative	Portfolio-based